

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/01/2020

Quiz 8: Section 3.11

Name (Print): _____

Name (Print): _____

Name (Print): _____

Section 3.11

5 pts

1. Sand falls from an overhead bin, accumulating in a conical pile with a radius that is always three times its height. Find a formula for $\frac{dh}{dt}$.

(Hint: The volume of a cone is $V = (1/3)\pi r^2 h$.)

Ⓐ $\frac{dh}{dt} = 3\pi h^3 \frac{dV}{dt}$

Ⓑ $\frac{dh}{dt} = \frac{9h}{\pi} \frac{dV}{dt}$

Ⓒ $\frac{dh}{dt} = \frac{V}{9\pi h^2}$

Ⓓ $\frac{dh}{dt} = \frac{dV}{dt} \frac{1}{9\pi h^2}$

5 pts

2. Sand falls from an overhead bin, accumulating in a conical pile with a radius that is always three times its height. If the sand falls from the bin at a rate of $120 \text{ ft}^3/\text{min}$, approximately how fast is the height of the sand pile changing when the pile is 11 ft high?

(Hint: The volume of a cone is $V = (1/3)\pi r^2 h$.)

Ⓐ 0.042 ft/min

Ⓑ 0.035 **ft/min**

Ⓒ 420.089 ft/min

Ⓓ 7.173 ft/min